

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations
May 2025

CB2102-1- FUNDAMENTALS OF COMPUTER NETWORKS
(For CB)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which type of service does the network layer provide to the transport layer in connection-oriented service?
A) Reliable end-to-end delivery
B) Datagram delivery
C) Broadcast delivery
D) File transfer
2. Which routing method allows routers to learn dynamically about their neighbors?
A) Broadcast Routing
B) Distance Vector Routing
C) Hierarchical Routing
D) Multicast Routing
3. What is a feature of multicast routing?
A) Sends packets to a single destination
B) Sends packets to multiple specific destinations
C) Requires tunneling
D) Routes packets only in wired networks
4. Which congestion control approach involves predefining limits for traffic?
A) Load shedding
B) Admission control
C) Traffic throttling
D) Packet shaping
5. Which of the following is an example of integrated services in QoS?
A) RSVP (Resource Reservation Protocol)
B) TCP/IP Protocol
C) Load Shedding
D) DNS Services
6. What is the role of a virtual circuit in the network layer?
A) Provides a direct physical link
B) Ensures guaranteed delivery of packets
C) Establishes a logical path for packet delivery
D) Handles congestion control only
7. Which algorithm relies on knowledge of the entire network topology?
A) Flooding
B) Link State Routing
C) Distance Vector Routing
D) Shortest Path First
8. Which routing method sends packets to a group of specific hosts?
A) Anycast Routing
B) Multicast Routing
C) Broadcast Routing
D) Unicast Routing
9. Which type of routing is best suited for mobile hosts?
A) Broadcast Routing
B) Routing in Ad hoc Networks
C) Hierarchical Routing
D) Distance Vector Routing
10. What is a key characteristic of connectionless services?
A) Guaranteed delivery of packets
B) Low latency in transmission
C) Packets may arrive out of order
D) Requires session establishment

Which congestion control technique drops packets to prevent overflow?

11. A) Load shedding B) Traffic shaping
C) Admission control D) Packet scheduling
12. **Which routing method sends packets to a group of specific hosts?**
A) Anycast Routing B) Multicast Routing
C) Broadcast Routing D) Unicast Routing
13. **What is the key difference between connection-oriented and connectionless services?**
A) Connection-oriented requires session establishment, connectionless does not B) Connection-oriented is faster than connectionless
C) Connection-oriented is less reliable than connectionless D) Connectionless requires acknowledgments for packets
14. **Which of the following is NOT a design issue of the network layer?**
A) Packet forwarding B) End-to-end delivery
C) Routing algorithms D) Error detection
15. **Which of the following is a proactive routing algorithm?**
A) Distance Vector Routing B) Flooding
C) On-Demand Routing D) Hierarchical Routing
16. **What does the term "traffic-aware routing" refer to?**
A) Routing based on user behavior B) Routing that adjusts dynamically to traffic conditions
C) Routing with fixed paths for each flow D) Routing using static tables
17. **Which of the following is an example of traffic shaping?**
A) Token Bucket Algorithm B) TCP Sliding Window
C) Link State Routing D) Admission Control
18. **What is the purpose of admission control in congestion control?**
A) Limiting traffic entry into the network B) Prioritizing packets
C) Increasing bandwidth D) Dividing packets into smaller sizes
19. **What does the term "datagram subnet" refer to?**
A) A connection-oriented network B) A network that uses connectionless service
C) A network with fixed routes D) A network with guaranteed delivery
20. **In Link State Routing, each router maintains information about:**
A) The entire network topology B) Only its neighbors
C) The shortest paths to all nodes D) The congestion status of the network

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the process of congestion control using traffic throttling and load shedding.	8	L*2	1	1
	b) Compare virtual circuit and datagram subnets and discuss their applications.	8	L3	2	2
2.	a) Describe the shortest path routing with an example. How is Dijkstra's algorithm used in this context?	8	L2	1	1
	b) Differentiate between connectionless and connection-oriented services in the network layer with examples.	8	L3	2	2
3.	a) Define the concept of store-and-forward packet switching. Explain how it is implemented in a network.	8	L2	1	1
	b) Analyze the working of distance vector routing with a step-by-step example.	8	L3	2	2

Unit – II

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| 4. a) | Define fragmentation in the network layer. Explain the process and how fragmented packets are reassembled. | 8 | L2 | 3 |
| b) | Discuss the structure of IPv4 and IPv6 addresses and list their differences. | 8 | L2 | 4 |
| 5. a) | Describe the role of OSPF and BGP in internetwork routing. | 8 | L2 | 3 |
| b) | Outline the services provided by the transport layer to the upper layers. | 8 | L2 | 4 |
| 6. a) | Explain the working of the User Datagram Protocol (UDP) with its key features and limitations. | 8 | L2 | 3 |
| b) | Write a short note on tunneling and its applications in network communication. | 8 | L2 | 3 |

Unit – III

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| 7. a) | Explain the TCP sliding window protocol with an example. | 8 | L2 | 5 |
| b) | Discuss the architecture of DNS and explain the types of records stored in DNS. | 8 | L2 | 5 |
| 8. a) | Explain the working of SMTP in electronic mail transfer and compare it with HTTP. | 8 | L2 | 5 |
| b) | Describe the TCP service model and explain how TCP manages reliable data transfer. | 8 | L2 | 5 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
